



NORTH SYDNEY BOYS HIGH SCHOOL

STAGE 5.1-5.3 MATHEMATICS

2011 Year 9 Final Examination

General instructions

- Working time – 90 minutes.
- Marks may be deducted for careless or poorly arranged work.
- Commence each new question on a new sheet.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

Class (please ✓)

- ☐ 9M1 – Mr Berry/Mr Lin
- ☐ 9M2 – Mr Lowe/Mr Lin
- ☐ 9M3 – Mr Lam
- ☐ 9M4 – Ms Ziazaris
- ☐ 9M5 – Mr Weiss/Mr Lin

NAME: # BOOKLETS USED:

Marker's use only.

QUESTION	1	2	3	4	5	6	7	8	Total	%
MARKS	$\overline{12}$	$\overline{12}$	$\overline{12}$	$\overline{12}$	$\overline{17}$	$\overline{12}$	$\overline{13}$	$\overline{10}$	$\overline{100}$	

Question 1	(12 Marks)	Commence a NEW page.	Marks
(a)	Simplify fully: $\frac{7ab}{4} - \frac{5ab}{4}$.		2
(b)	Solve the equation $17 + 6y = 35$.		2
(c)	Expand and simplify:		
	i. $(x - 8)(x + 2)$.		2
	ii. $(m^2 + 3n)^2$.		2
(d)	Fully factorise:		
	i. $4a^2 + 10a$.		2
	ii. $x^2 - 36$.		2

Question 2	(12 Marks)	Commence a NEW page.	Marks
(a)	Fully simplify $x^{2a-1} \times x^{a+3}$.		2
(b)	Divide \$2 247 in the ratio 3 : 2 : 1.		2
(c)	Solve:		
	i. $\frac{m+6}{3} = \frac{2m+4}{4}$.		2
	ii. $\frac{4}{y-3} + 2 = 1$.		2
(d)	Solve $2x^2 + 3x - 3 = 0$, giving your solutions as exact values.		2
(e)	A light year is the distance light travels in a year.		2

The speed of light is 2.998×10^5 km per second and there are 365.25 days in a year.
Find the number of kilometres in a light year in scientific notation, correct to 3 significant figures.

Question 3 (12 Marks)

Commence a NEW page.

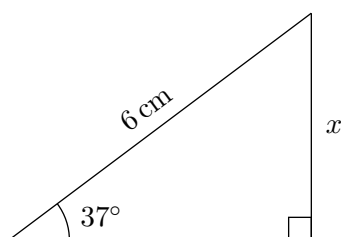
Marks

- (a) Mrs Short has a taxable income of \$86 770. Using the tax table below, find her tax payable. **2**

Taxable income	Tax on this income
\$0 – \$6 000	Nil
\$6 001 – \$37 000	15c for each \$1 over \$6 000
\$37 001 – \$80 000	\$4 650 plus 30c for each \$1 over \$37 000
\$80 001 – \$180 000	\$17 550 plus 37c for each \$1 over \$80 000
\$180 001 and over	\$54 550 plus 45c for each \$1 over \$180 000

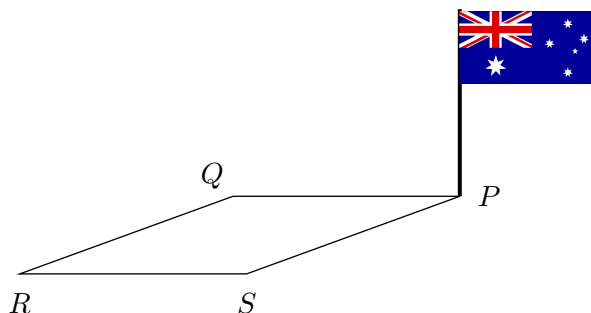
- (b) Use your calculator to find $\tan 78.4^\circ$ correct to 3 decimal places. **1**

- (c) Find x correct to three significant figures. **2**



- (d) If $\sin \theta = -\frac{1}{\sqrt{6}}$ and $\tan \theta > 0$, find the exact value of $\cos \theta$. **3**

- (e) A flagpole stands at one corner P of a level square field $PQRS$, each side being 120 m long.



If the angle of elevation to the top of the pole from Q is $13^\circ 19'$, find:

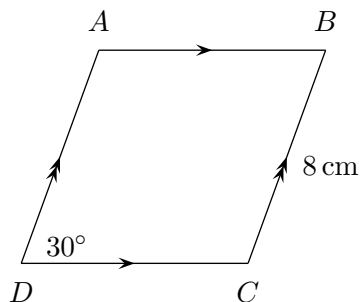
- i. the height of the flagpole, correct to the nearest centimetre. **1**
- ii. its angle of elevation from R . **3**

Question 4 (12 Marks) Commence a NEW page. **Marks**

(a) Find the radius of a sphere with volume $1\,220\text{ cm}^3$ correct to 1 decimal place. **2**

(b) A cone has a diameter of 16 cm and perpendicular height of 15 cm. Calculate the curved surface area of the cone in terms of π . **2**

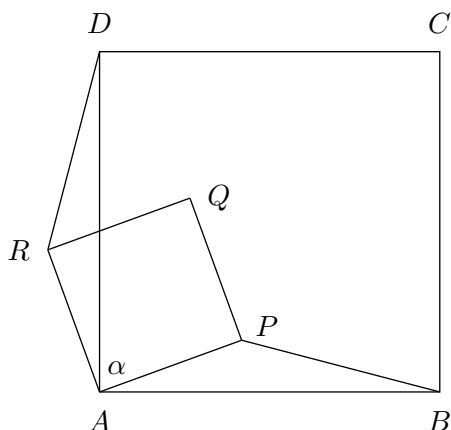
(c) In the diagram, $ABCD$ is a parallelogram with $\angle ADC = 30^\circ$ and $BC = 8\text{ cm}$.



i. Find the size of $\angle ABC$, giving a reason. **2**

ii. If the diagonals AC and BD are perpendicular, find the perimeter $ABCD$. **2**
Give reasons.

(d) $ABCD$ and $APQR$ are squares and $\angle DAP = \alpha$.



i. Prove $\triangle DRA \equiv \triangle ABP$. **3**

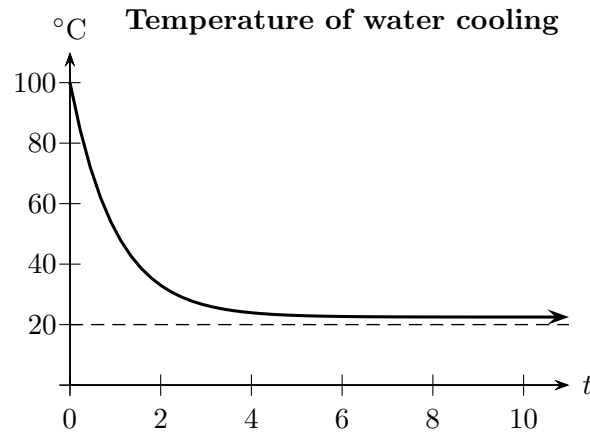
ii. Hence or otherwise, given a reason why $BP = DR$. **1**

Question 5 (17 Marks)

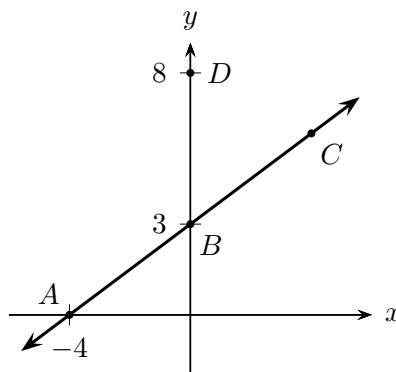
Commence a NEW page.

Marks

- (a) Sketch the region represented simultaneously by the inequalities $x + y \leq 1$ and $y \geq 2x$. **3**
- (b) The graph shows the temperature of some hot water which is cooling over time. Use this graph to answer the following questions.



- i. Was the water cooling more rapidly at 2 hours or at 8 hours? Give reasons for your answer. **2**
- ii. As time passes, the temperature of the water approaches the temperature of the room. Write down the temperature of the room. **1**
- (c) The diagram shows line ℓ cutting the x axis at $A(-4, 0)$ and the y axis at $B(0, 3)$. The point D has coordinates $(0, 8)$.



- i. Find the gradient of the line ℓ . **1**
- ii. Show that the line ℓ has equation $3x - 4y + 12 = 0$. **2**
- iii. The point C lies on ℓ and B is the midpoint of AC . Prove that the coordinates of C are $(4, 6)$. **2**
- iv. Show that the exact length of AD is $4\sqrt{5}$ units. **2**
- v. Show that AD is perpendicular to DC . **2**
- vi. Find the area of $\triangle ADC$. **2**

Question 6 (12 Marks)	Commence a NEW page.	Marks
(a) Simplify fully $\sqrt{45} + \sqrt{5}$.		2
(b) Express $\frac{\sqrt{3}}{\sqrt{3}-1}$ as the simplest fraction with a rational denominator.		2
(c) Solve the following pair of simultaneous equations: $\begin{cases} y = x - 10 \\ x - 3y = 6 \end{cases}$.		3
(d) Solve $x + \frac{10}{x} = 7$.		2
(e) Fully simplify the following expression: $\frac{1}{x^2 - x - 6} + \frac{4}{x^2 + 8x + 12}$.		3

Question 7 (13 Marks)	Commence a NEW page.	Marks
(a) Boxes of soap powder are each labelled as having a mass of 1 kg. The mass of these packets is distributed normally with a mean of 1.02 kg and a standard deviation of 0.01 kg.		2
What percentage of boxes have an actual mass between 1 kg and 1.03 kg?		
(b) Consider the following scores: 4, 6, 7, 8, 9, 10, 10, 11, 11, 12, 16, 17, 18, 19, 20		
i. Find the standard deviation.		1
ii. What is the median?		1
iii. What is the lower quartile Q_1 ?		1
iv. What is the upper quartile Q_3 ?		1
v. Find the interquartile range.		1
vi. Draw a box-and-whisker plot showing the above information.		2
(c) Vincent travels to school every day by train. He has found that the probability the train is late by five or more minutes is 7% on Mondays and 13% on Tuesdays.		
i. Draw a probability tree diagram for the late trains on Monday and Tuesday.		2
ii. Find the probability that the train is late by five minutes or more on <i>either</i> day, but not both.		2

Question 8 (10 Marks)

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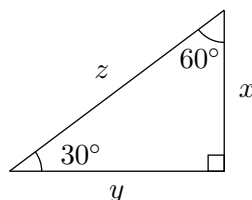
Marks

- (a) Find
- k
- in terms of
- n
- given

3

$$x^k = \sqrt{\frac{x^{n+1} \times (x^2)^{n+2}}{\sqrt{x} \times x^{n+\frac{1}{2}}}}$$

- (b) A piece of wire of length 20 cm is bent in the shape of a triangle with interior angles
- 30°
- ,
- 60°
- and
- 90°
- .



- i. Show that $y = \sqrt{3}x$ and $z = 2x$. **2**
- ii. Hence or otherwise, find the length of the hypotenuse, giving your answer in surd form with a rational denominator. **2**
- (c) Sammi thinks of a two-digit number. The sum of the digits is 8. If she reverses the digits the new number is 36 greater than her original number. What was Sammi's original number? **3**

End of paper.

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Suggested Solutions

Question 1

(a) (2 marks)

✓ [1] for $\frac{2ab}{4}$.

✓ [1] for $\frac{ab}{2}$.

$$\frac{7ab}{4} - \frac{5ab}{4} = \frac{\cancel{2}ab}{\cancel{4}_2} = \frac{ab}{2}$$

(b) (2 marks)

✓ [-1] for each error.

$$\begin{array}{r} 17 \\ -17 \end{array} + 6y = \begin{array}{r} 35 \\ -17 \end{array}$$

$$6y = 18$$

$$y = 3$$

(c) i. (2 marks)

✓ [-1] for each error.

$$(x - 8)(x + 2) = x^2 - 6x - 16$$

ii. (2 marks)

✓ [-1] for each error.

✓ [0] for $m^4 + 9n^2$.

$$(m^2 + 3n)^2 = m^4 + 6m^2n + 9n^2$$

(d) i. (2 marks)

✓ [-1] for each error.

$$4a^2 + 10a = 2a(2a + 5)$$

ii. (2 marks)

✓ [-1] for each error.

$$x^2 - 36 = (x - 6)(x + 6)$$

Question 2

(a) (2 marks)

✓ [-1] for each error.

$$x^{2a-1} \times x^{a+3} = x^{2a+a-1+3} = x^{3a+2}$$

(b) (2 marks)

✓ [1] for finding 1 part.

✓ [1] for final answer.

$$1 \text{ part} = \frac{2\,247}{6} = 374.5$$

$$\therefore 3 : 2 : 1 = \$1\,123.5 : \$749 : \$374.5$$

(c) i. (2 marks)

✓ [1] for correctly cross multiplying.

✓ [1] for final answer.

$$\frac{m+6}{3} = \frac{2m+4}{4} = \frac{\cancel{2}(m+2)}{\cancel{4}_2}$$

$$2(m+6) = 3(m+2)$$

$$\begin{array}{cccc} 2m & + & 12 & = & 3m & + & 6 \\ -2m & & -6 & & -2m & & -6 \end{array}$$

$$\therefore m = 6$$

ii. (2 marks)

✓ [-1] for each error.

$$\frac{4}{y-3} + \frac{2}{-2} = \frac{1}{-2}$$

$$\frac{4}{y-3} = \frac{-1}{(\div -1)} \quad \begin{array}{l} (\times (y-3)) \\ (\div -1) \end{array}$$

$$\begin{array}{ccc} -4 & = & y - 3 \\ +3 & & +3 \end{array}$$

$$y = -1$$

(d) (2 marks)

✓ [1] for recalling quadratic formula.

✓ [1] for (both) final answers.

$$2x^2 + 3x - 3 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-3 \pm \sqrt{9 - 4(2)(-3)}}{2 \times 2}$$

$$= \frac{-3 \pm \sqrt{33}}{4}$$

(e) (2 marks)

✓ [1] for multiplying by $60 \times 60 \times 24 \times 365.25$.

✓ [1] for final answer.

$$\frac{2.998 \times 10^5 \text{ km}}{1 \text{ sec}} \frac{\times 60 \times 60 \times 24 \times 365.25}{\times 60 \times 60 \times 24 \times 365.25}$$

$$= \frac{9.46 \times 10^{12} \text{ km}}{1 \text{ year}}$$

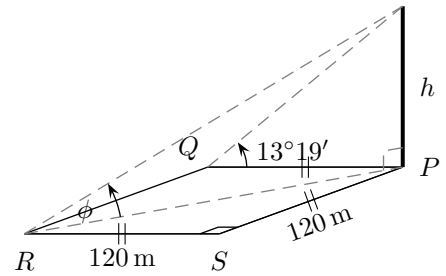
Question 3

(e) i. (1 mark)

(a) (2 marks)

- ✓ [1] for tax payable on marginal rate.
- ✓ [1] for final answer.

$$17\,550 + 0.37(86\,770 - 80\,000) \\ = \$20\,054.90$$



(b) (1 mark)

$$\tan 78.4^\circ = 4.872$$

$$\frac{h}{120} = \tan 13^\circ 19' \\ h = 120 \tan 13^\circ 19' = 28.4 \text{ m}$$

(c) (2 marks)

- ✓ [1] for identifying sin ratio.
- ✓ [1] for final answer.

$$\frac{x}{6} = \sin 37^\circ \\ x = 6 \sin 37^\circ = 3.61$$

ii. (3 marks)

- ✓ [1] for finding RP .
- ✓ [1] for $\frac{h}{PR} = \tan \phi$.
- ✓ [1] for final answer.

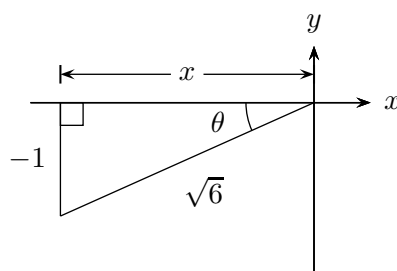
In $\triangle PSR$,

$$PR^2 = PS^2 + RS^2 = 120^2 + 120^2 \\ = 2 \times 120^2 \\ \therefore PR = 120\sqrt{2}$$

$$\tan \phi = \frac{h}{PR} \\ = \frac{28.4}{120\sqrt{2}} \\ = 0.167 \dots \\ \therefore \phi = 9^\circ 30'$$

(d) (3 marks)

- ✓ [1] for identifying 3rd quadrant.
- ✓ [1] for finding missing side.
- ✓ [1] for final answer.



$$x^2 + 1^2 = (\sqrt{6})^2 \\ x^2 = 5 \\ \therefore x = \sqrt{5} \\ \therefore \cos \theta = -\frac{\sqrt{5}}{\sqrt{6}} = -\sqrt{\frac{5}{6}}$$

Question 4

(a) (2 marks)

- ✓ [1] for recalling formula for volume of sphere.
- ✓ [1] for final answer.

$$V = \frac{4}{3}\pi r^3$$

$$1\,220 = \frac{4}{3}\pi r^3$$

$$r^3 = \frac{1\,220 \times 3}{4}$$

$$\therefore r \approx 6.628 \text{ cm}$$

(b) (2 marks)

- ✓ [1] for correctly calculating slanted height.
- ✓ [1] for final answer.

$$s^2 = 8^2 + 15^2 = 289$$

$$\therefore s = 17$$

$$SA = \pi rs = \pi \times 8 \times 17$$

$$= 136\pi$$

(c) i. (2 marks)

- ✓ [1] for final answer.
- ✓ [1] for correct reason.

$$\angle ABC = 30^\circ$$

(opposite $\angle$ in parallelogram)

ii. (2 marks)

- ✓ [1] for final answer.
- ✓ [1] for correct reason.

- $P = 8 \times 4 = 32 \text{ cm.}$

(Diagonals of a parallelogram bisect each other, and since they are $\perp$, $\therefore$ the parallelogram is a rhombus. Sides of a rhombus are equal.)

(d) i. (3 marks)

- ✓ [-1] for each missing/incorrect step in the proof.
- $AB = AD$
(sides of a square $ABCD$)

- $\angle PAB = \angle RAD$
($\angle$ in square; complementary to α in squares $ABCD$ and $APQR$ respectively)

- $AP = AR$
(sides of a square $APQR$)

$\therefore \triangle ABP \equiv \triangle DRA$ (SAS)

ii. (1 mark)

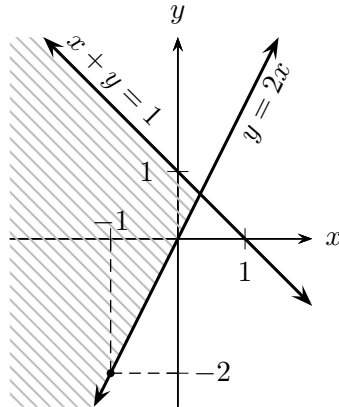
$$BP = DR$$

(corresponding sides of congruent $\triangle ABP$ & $\triangle DRA$)

Question 5

(a) (3 marks)

- ✓ [1] for sketching $x + y = 1$ correctly.
- ✓ [1] for sketching $y = 2x$ correctly.
- ✓ [1] for correctly shaded region.



(b) i. (2 marks)

- ✓ [1] for 2 hrs.
- ✓ [1] for reason.

2 hrs – graph is more steep.

ii. (1 mark) 20°C

(c) i. (1 mark) $m = \frac{3-0}{0-(-4)} = \frac{3}{4}$

ii. (2 marks)

- ✓ [1] for each of m and b .

$$y = \frac{3}{4}x + \frac{3}{4}$$

$$4y = 3x + 12$$

$$\therefore 3x - 4y + 12 = 0$$

(use gradient from previous part,
read off y intercept)

iii. (2 marks)

- ✓ [1] for using midpoint formula correctly
- ✓ [1] for solving for x and y .
- ✓ [0] for simply restating question etc.

$$B(0, 3) = \left(\frac{-4+x}{2}, \frac{0+y}{2} \right)$$

$$\frac{-4+x}{2} = 0 \quad \left| \quad \frac{0+y}{2} = 3 \right.$$

$$\begin{array}{l} -4+x=0 \\ x=4 \end{array} \quad \left. \begin{array}{l} y=6 \end{array} \right\} \therefore C(4, 6)$$

iv. (2 marks)

$$A(-4, 0) \quad D(0, 8)$$

$$AD = \sqrt{(-4)^2 + 8^2}$$

$$= \sqrt{4^2 + 8^2}$$

$$= \sqrt{16 + 64} = \sqrt{80}$$

$$= 4\sqrt{5}$$

v. (2 marks)

- ✓ [-1] for each error.

$$m_{AD} = \frac{8-0}{0-(-4)} = \frac{8}{4} = 2$$

$$m_{DC} = \frac{8-6}{0-4} = \frac{2}{-4} = -\frac{1}{2}$$

$$m_{AD} \times m_{DC} = -1$$

$$\therefore AD \perp DC$$

vi. (2 marks)

- ✓ [1] for length of DC .
- ✓ [1] for final answer.

$$DC = \sqrt{(4-0)^2 + (6-8)^2}$$

$$= \sqrt{4^2 + 2^2} = \sqrt{20}$$

$$= 2\sqrt{5}$$

$$A_{\triangle ADC} = \frac{1}{2}bh$$

$$= \frac{1}{2} \times 4\sqrt{5} \times 2\sqrt{5}$$

$$= 4 \times 5 = 20 \text{ m}^2$$

Question 6

(a) (2 marks)

- ✓ [1] for simplifying $\sqrt{45}$.
- ✓ [1] for final answer.

$$\sqrt{45} + \sqrt{5} = 3\sqrt{5} + \sqrt{5} = 4\sqrt{5}$$

(b) (2 marks)

- ✓ [1] for multiplying by conjugate in numerator & denominator.
- ✓ [1] for final answer.

$$\begin{aligned} \frac{\sqrt{3}}{\sqrt{3}-1} \frac{\times(\sqrt{3}+1)}{\times(\sqrt{3}+1)} &= \frac{\sqrt{3}(\sqrt{3}+1)}{3-1} \\ &= \frac{\sqrt{3}(\sqrt{3}+1)}{2} \end{aligned}$$

(c) (3 marks)

- ✓ [1] for correct method.
- ✓ [1] for each of x and y .

$$\begin{cases} y = x - 10 & (1) \\ x - 3y = 6 & (2) \end{cases}$$

Substitute (1) into (2)

$$\begin{aligned} x - 3(x - 10) &= 6 \\ -2x + 30 &= 6 \\ 2x &= 24 \\ x &= 12 \\ \therefore y &= 2 \end{aligned}$$

(d) (2 marks)

- ✓ [1] for each correct root.

$$\begin{aligned} x + \frac{10}{x} &= 7 \\ \times x & \quad \times x \\ x^2 + 10 &= 7x \\ x^2 - 7x + 10 &= 0 \\ (x - 5)(x - 2) &= 0 \\ \therefore x &= 2, 5 \end{aligned}$$

(e) (3 marks)

- ✓ [1] for correct factorisations in denominators.
- ✓ [1] for correct common denominator.
- ✓ [1] for final answer.

$$\begin{aligned} &\frac{1}{x^2 - x - 6} + \frac{4}{x^2 + 8x + 12} \\ &= \frac{1}{(x-3)(x+2)} + \frac{4}{(x+6)(x+2)} \\ &= \frac{(x+6) + 4(x-3)}{(x-3)(x+2)(x+6)} \\ &= \frac{x+6+4x-12}{(x-3)(x+2)(x+6)} \\ &= \frac{5x-6}{(x-3)(x+2)(x+6)} \end{aligned}$$

Question 7

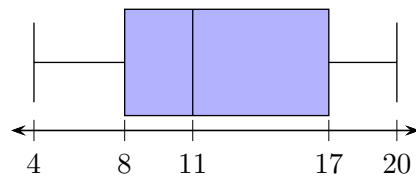
(a) (2 marks)

- ✓ [1] for identifying $\mu - 2\sigma$ and $\mu + \sigma$ locations.
 ✓ [1] for final answer.

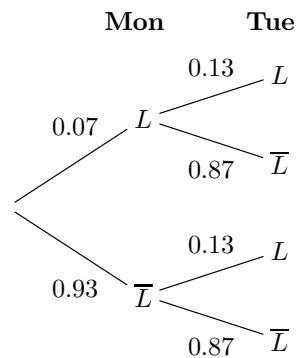
$$P(1 < x < 1.03) = 0.475 + 0.34 = 0.815$$

(b) i. (1 mark) $\sigma = 4.83$ ii. (1 mark) $\tilde{x} = 11$.iii. (1 mark) $Q_1 = 8$ iv. (1 mark) $Q_3 = 17$ v. (1 mark) $IQR = 9$

vi. (2 marks)



(c) i. (2 marks)



ii. (2 marks)

$$\begin{aligned}
 P(\overline{L}\overline{L}, \overline{L}L) &= (0.07 \times 0.87) + (0.93 \times 0.13) \\
 &= 0.1818
 \end{aligned}$$

Question 8

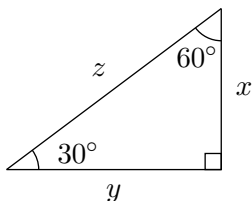
(a) (3 marks)

- ✓ [-1] for each error.
- ✓ [3] if students have simplified all correctly down to x^{n+2} .

$$\begin{aligned}
 x^k &= \sqrt{\frac{x^{n+1} \times (x^2)^{n+2}}{\sqrt{x} \times x^{n+\frac{1}{2}}}} \\
 &= \left(\frac{x^{n+1} \times x^{2n+4}}{x^{\frac{1}{2}} \times x^{n+\frac{1}{2}}} \right)^{\frac{1}{2}} \\
 &= \left(\frac{x^{3n+5}}{x^{n+1}} \right)^{\frac{1}{2}} \\
 &= (x^{2n+4})^{\frac{1}{2}} \\
 &= x^{n+2} \\
 \therefore k &= n+2
 \end{aligned}$$

(b) i. (2 marks)

- ✓ [1] for proving each relationship correctly.



$$\begin{aligned}
 \tan 60^\circ &= \sqrt{3} = \frac{y}{x} \\
 \therefore y &= \sqrt{3}x
 \end{aligned}$$

Also,

$$\begin{aligned}
 \cos 60^\circ &= \frac{1}{2} = \frac{x}{z} \\
 \therefore z &= 2x
 \end{aligned}$$

ii. (2 marks)

- ✓ [1] for obtaining $x = \frac{20}{3+\sqrt{3}}$.
- ✓ [1] for final answer with rationalised denominator.

$$\begin{aligned}
 x + y + z &= 20 \\
 x + \sqrt{3}x + 2x &= 20 \\
 x(1 + \sqrt{3} + 2) &= 20 \\
 x &= \frac{20}{3 + \sqrt{3}}
 \end{aligned}$$

As z is the hypotenuse,

$$\begin{aligned}
 z = 2x &= \frac{40}{3 + \sqrt{3}} \times \frac{(3-\sqrt{3})}{(3-\sqrt{3})} \\
 &= \frac{40(3-\sqrt{3})}{9-3} \\
 &= \frac{20(3-\sqrt{3})}{3}
 \end{aligned}$$

(c) (3 marks)

- ✓ [1] for observing the original number is $10x + y$, if x is the first digit and y is the second.
- ✓ [1] for correctly constructing second worded equation.
- ✓ [1] for obtaining $x = 2$.
- Let the number have digits x and y , x being the first digit, i.e. $(x \times 10^1) + (y \times 10^0) = 10x + y$.
- Switching the digits results in $10y + x$.

$$\begin{cases} x + y = 8 & (1) \\ 10y + x = 36 + (10x + y) & (2) \end{cases}$$

Simplifying (2),

$$\begin{aligned}
 10y + x &= 36 + 10x + y \\
 9x + 36 &= 9y \\
 \therefore x + 4 &= y
 \end{aligned}$$

Substitute to (1),

$$\begin{aligned}
 x + (x + 4) &= 8 \\
 2x + 4 &= 8 \\
 \therefore x &= 2 \therefore y = 6
 \end{aligned}$$

The original number is 26.